

Storms and hail

Deep cyclones can create very high winds that cause massive damage. They also bring widespread rain, but the biggest cloudbursts are produced by giant cumulonimbus clouds that are built up by intense heat and evaporation. Air currents inside the clouds create hail and thunderstorms.

FRONTAL STORMS

When warm and cold air masses meet at the polar fronts, deep cyclones can build up over the oceans, especially in early autumn when the sea is still warm. The warm, moist, rising air at the center of the cyclone acts like a vacuum cleaner, sucking in low-level air. This creates powerful winds that swirl toward the middle of the cyclone. When one of these major storms moves over land, it can be powerful enough to bring down trees.



CLOUDBURSTS

Tall cumulonimbus clouds suck up vast amounts of moisture. The water droplets grow larger and larger until they fall from the cloud. A single cloudburst can release up to 300,000 tons (275,00 metric tons) of water, deluging the land and causing flash flooding.



HAILSTORMS

Ice crystals form near the tops of cumulonimbus clouds. They then fall through the cloud, picking up water. Powerful updrafts hurl the crystals back up, and the water freezes onto them, making them bigger. They do this many times until they are heavy enough to fall to the ground as hailstones, which can become as large as tennis balls.